nVIDIA Inc.

nVidia remained relatively low-key until the late 1997-98 period, when it launched its line of RIVA PC graphics processors.

nVidia's original graphics card—the NV1—was released in 1995. It had an integrated playback-only sound card. It struggled in a market full of several competing proprietary standards.

Market interest in the NV1 ended when Microsoft announced the DirectX specifications. NV1 development then continued as the NV2 project, funded by investment from Sega. Sega hoped an integrated sound and graphics chip would cut the manufacturing cost of their next console. nVidia thus abandoned proprietary interfaces, and sought to fully support DirectX. It dropped multimedia functionality.

The year 1999 saw the release of the GeForce 256 (NV10), most notably bringing on-board transformation and lighting. The GF256 ran at 120 MHz, and was implemented with advanced video acceleration, motion compensation, and had four-pixel pipelines. When combined with DDR memory support, nVidia's technology was the performance leader. Basking in the success of its products, nVidia won the contract to develop graphics hardware for the Xbox.

All this was too much for 3Dfx, whose Voodoo 5 had been delayed; nVidia purchased 3Dfx—primarily for the intellectual property that was in dispute at the time. But nVidia also acquired anti-aliasing expertise and a lot of engineers.

Subsequent problems between nVidia and Microsoft meant nVidia was not consulted when the DirectX 9 specification was drawn up. ATi then designed its 9700 to fit DirectX specifications. Rendering colour support was limited to 24 bits, and shader performance was emphasised. In contrast, nVidia's cards offered 16- and 32-bit modes, offering either poor visual quality or slow performance. The 32-bit support made them much more expensive to manufacture, and shader performance was often only half or less the speed provided by ATi.